

JEE MAIN REVISION MASTER: DUAL NATURE OF RADIATION SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 207 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Photoelectric Effect & Einstein's Equation	129 PYQs	Photoelectric
Very High	De Broglie Hypothesis & Matter Waves	78 PYQs	Matter Waves

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Photoelectric Effect & Einstein's Equation	$E = h\nu = \phi_0 + K_{\text{max}}$, $K_{\text{max}} = eV_s = h(\nu - \nu_0) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$
De Broglie Hypothesis & Matter Waves	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$, $\lambda_{\text{electron}} \approx \frac{1.227}{\sqrt{V}} \text{ nm}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Photoelectric Effect & Einstein's Equation (Very High Yield — 129 PYQs)

Core Principle: Quantum nature of light, work function ϕ_0 , threshold frequency ν_0 , and stopping potential V_s dependence on frequency.

■ **Caution / JEE Pitfall:** Stopping potential depends solely on FREQUENCY and work function; it is completely independent of intensity of incident light!

■ **Speed Trick:** Slope of stopping potential vs frequency curve is always universal: Slope = h/e .

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V. The work function of the used ..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "A light wave described by $E=60[\sin(3 \times 10^{15}t)+\sin(12 \times 10^{15}t)]$ (in SI units) falls on a metal surface of work function 2.8..." [Ans: D]
- [2026 Evening Shift] Q#2 (MCQ): " K_1 and K_2 be the maximum kinetic energies of photoelectrons emitted from a surface of a given material for the light of wavelength..." [Ans: D]

3.2 De Broglie Hypothesis & Matter Waves (Very High Yield — 78 PYQs)

Core Principle: Dual wave-particle nature of moving particles, de Broglie wavelength under electrostatic acceleration V , and thermal gas particles.

■■ **Caution / JEE Pitfall:** For neutral particles (neutrons), de Broglie wavelength is $\lambda = \frac{h}{\sqrt{3mkT}}$, where k is Boltzmann constant.

■ **Speed Trick:** For proton vs alpha particle accelerated through same voltage V : $(\lambda_p / \lambda_\alpha) = \sqrt{(m_\alpha q_\alpha / m_p q_p)} = \sqrt{(4 \times 2)} = 2\sqrt{2}$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons /s . The region of an electromagnetic spectrum to wh..." **[Ans: D]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____. (Given : $h = 6.63 \times 10^{-34}$..." **[Ans: D]**
- **[2026 Morning Shift] Q#1 (Numerical):** "The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n:1$. T..." **[Ans: 2]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons /s . The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to ____ ■ (Take $h = 6.6 \times 10^{-34}$ J.s and $c = 3 \times 10^8$ m/s).

- (A) Microwave
- (B) Infrared
- (C) Visible
- (D) Ultraviolet **[Correct]**

Correct Answer: D

Solution Analysis: The power (P) of a light source is the total energy emitted per unit time. If the source emits n photons per second, and each photon has an energy E , then the power is: $P = n \times E$ The given power of the source is $P = 15 \text{ kW} = 15 \times 10^3 \text{ W}$ and the number of photons per second is $n = 2.5 \times 10^{22} \text{ s}^{-1}$ $E = (P / n) \Rightarrow E = \frac{15 \times 10^3}{2.5 \times 10^{22}} \Rightarrow E = 6 \times 10^{-19} \text{ J}$ The energy of a photon is related to its wavelength by the formula: $E = (hc / \lambda) \Rightarrow \lambda = (hc / E)$ Substituting the constants and energy, The Pl

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V . The work function of the used metal in the experiment is $\alpha \times 10^{-19} \text{ J}$. The value of α is ____ . ($h=6.62 \times 10^{-34} \text{ Js}$, $e=1.6 \times 10^{-19} \text{ C}$ and $c=3 \times 10^8 \text{ m/s}$)

- (A) 3.68
- (B) 4.68
- (C) 5.68 **[Correct]**
- (D) 2.68

Correct Answer: C

Solution Analysis: If the energy of an incident photon is E then it is equal to the energy required to overcome the work function (ϕ) of the metal and the maximum kinetic energy (K_{max}) of the emitted electron. $E = \phi + K_{\text{max}}$ Therefore, $\phi = E - K_{\text{max}}$. Energy is given by $E = (hc / \lambda)$ where the Planck's constant $h = 6.62 \times 10^{-34} \text{ Js}$, the speed of light $c = 3 \times 10^8 \text{ m/s}$, and the wavelength of light is $\lambda = 331 \text{ nm} = 331 \times 10^{-9} \text{ m}$: Putting the values, $E = \frac{(6.62 \times 10^{-34}) \times (3 \times 10^8)}{331 \times 10^{-9}} \Rightarrow E = \frac{19.86}{1} \times 10^{-19} \text{ J}$